Memory allocation

Taking into account the hardware requirements, as well as practical experience in running the system, the memory for each element of the simulated environment was adjusted as shown in the table below. This reflects the observation that the Splunk Enterprise guest and the host operating system required more memory than originally anticipated, whereas pfSense and Caldera server machines were able to perform adequately with less RAM.

Table 1: Memory requirements and RAM allocation for the APT29 attack scenario components

Infrastructure component	Operating system	Minimum memory requirements	RAM allocation
Splunk Enterprise guest	Ubuntu 24.04	12 GB	15.0 GB
pfSense guest	FreeBSD	1 GB	1 GB
Domain controller guest	Windows Server 2016	2 GB	5 GB
Redirector guest	Ubuntu 16.10	0.384 GB	0.5 GB
Caldera server guest	Kali Linux	2 GB	2 GB
Windows client 1 guest	Windows 10 1903 version	2 GB	2 GB
Windows client 2 guest	Windows 10 1903 version	2 GB	2 GB
Linux host	Ubuntu 24.04	2 GB	4.5 GB